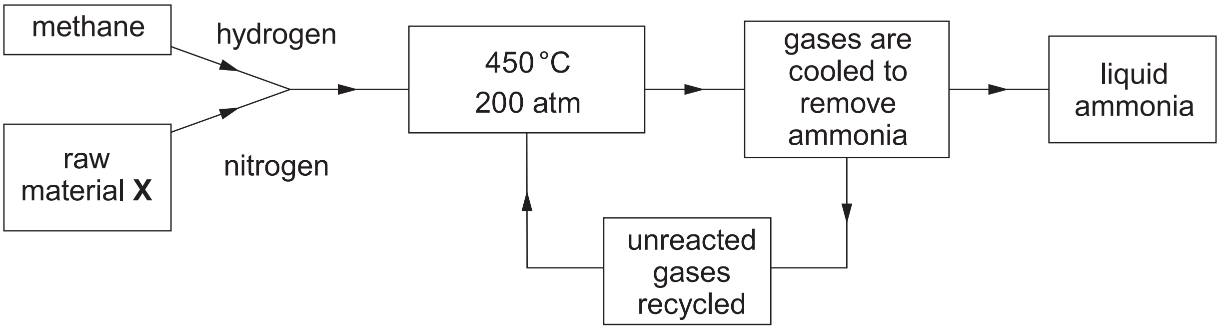
Examiner only

9. (a) The diagram outlines the manufacture of ammonia by the Haber process.



(i) Name the raw material **X**. .... [1]

(ii) The pressure used in the Haber process is 200 atm. State why a higher pressure is **not** used. [1]

....

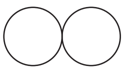
(iii) At 450 °C, the reaction is very slow. Iron is used in the process to speed up the reaction. Give the name for a substance used to speed up a chemical reaction. [1]

....

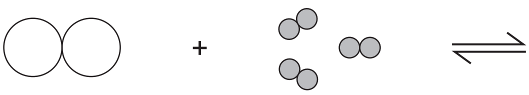
(iv) The reaction between nitrogen and hydrogen is represented by the equation below.

$$N_2 \quad + \quad 3H_2 \quad \rightleftharpoons \quad 2NH_3$$

Complete the equation below using the key: [2]

 nitrogen gas, N_2

 hydrogen gas, H_2



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(b) One of the main uses of ammonia is in the manufacture of fertilisers.

The table shows the results obtained when tests were carried out on three different fertilisers **A**, **B** and **C**.

Fertiliser	Test for positive ion	Test for negative ion
A	On adding sodium hydroxide solution and warming, a pungent smelling gas is formed which turns red litmus blue	On adding barium chloride solution a white precipitate forms
B	Lilac flame test	On adding silver nitrate solution a white precipitate forms
C	On adding sodium hydroxide solution and warming, a pungent smelling gas is formed which turns red litmus blue	On adding silver nitrate solution a white precipitate forms

Give the **letter** of the fertiliser which is ammonium sulfate. [1]

Letter ....

(c) Ammonia reacts with chlorine to form nitrogen and hydrogen chloride.

Complete the balancing of the equation for this reaction. [1]

$$2NH_3 + 3Cl_2 \longrightarrow N_2 + \boxed{}\,HCl$$

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 (3410U20-1) **Turn over.**